

Db2, IDAA, IMS & Safeguarded Copy: Resiliency Evolution and Integration

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Agenda

- Cyber Resiliency
- Db2, IDAA, IDAA & Safeguarded Copy Recovery
- Roll Forward Recovery Steps
- Continuous Data Validation
- Forensic Analysis
- Surgical Recovery
- Questions



Cyber Security and Cyber Resilience

Cyber Security

Cyber security is about prevention; it's about trying to keep the bad actors out of your environment

Cyber Resilience

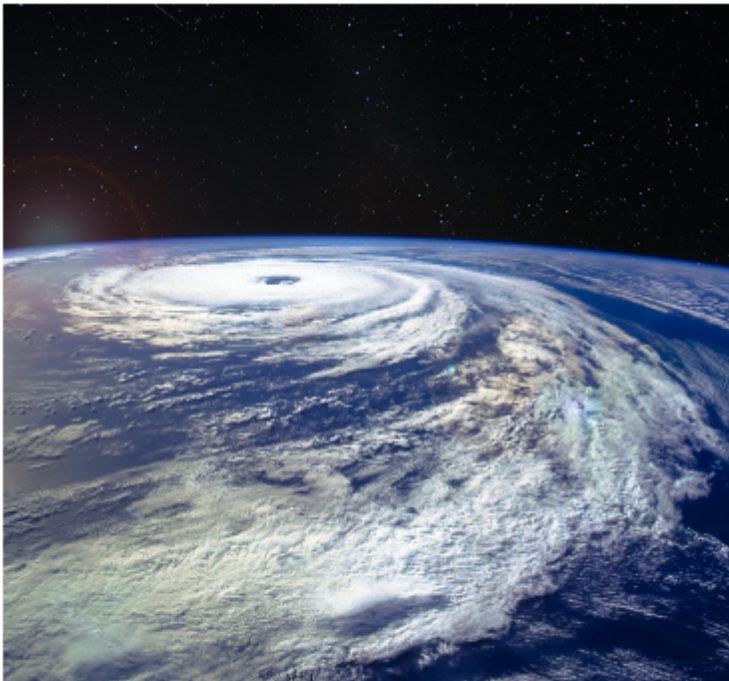
Cyber resilience is about an organizations ability to continue operations and rapidly recover despite a cyber-incident

Organizations need to be both cyber secure and cyber resilient



Cyber Security and Cyber Resilience ...

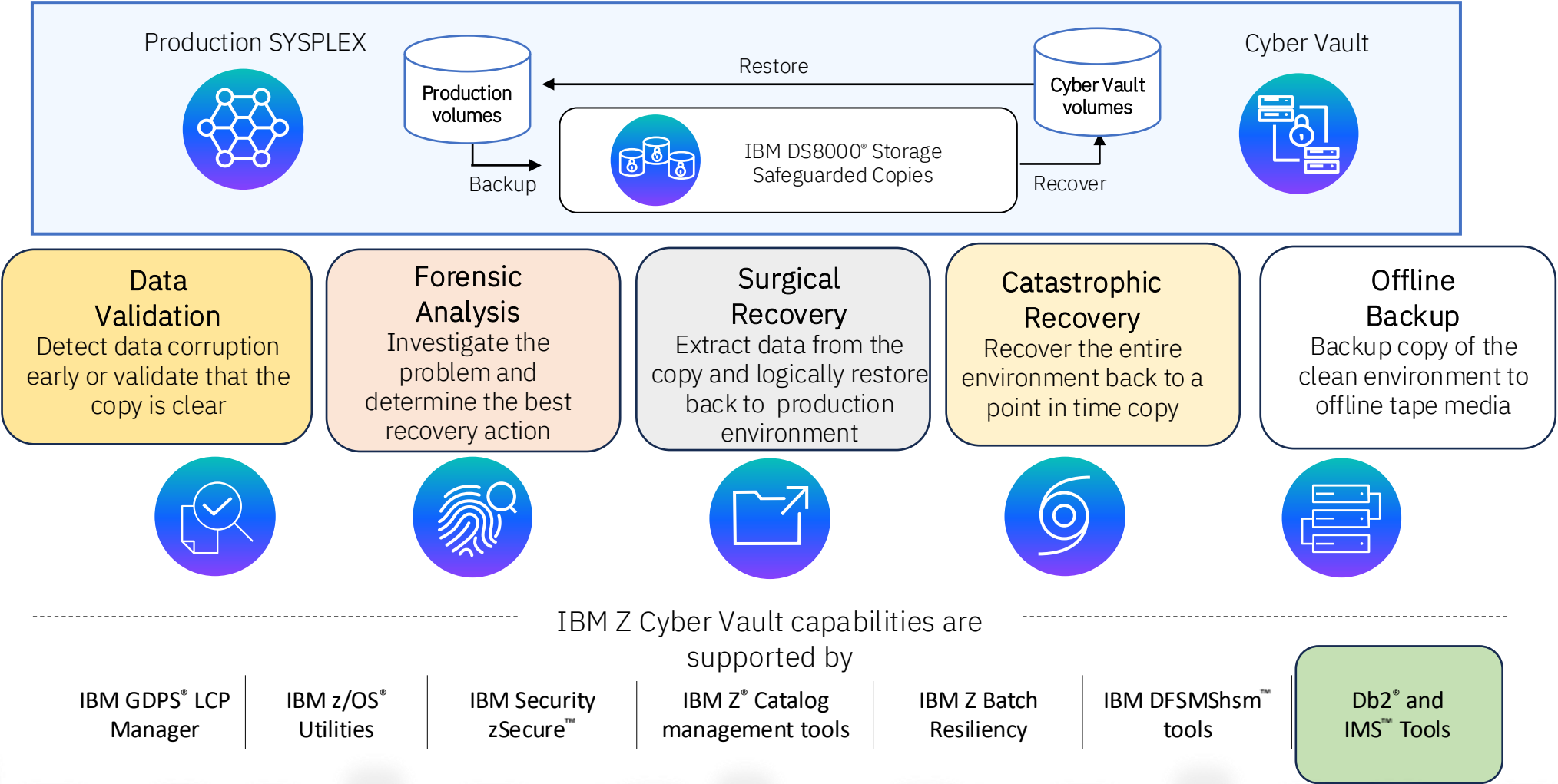
Traditional resiliency solutions
will not protect you from cyber
attack



	Traditional resiliency	What is required
Replication	Data is being replicated continuously but logical errors are also replicated instantaneously	Scheduled point in time copies stored in an isolated, secure location
Error detection	Immediate detection of system and application outages	Regular data analytics on point in time copies to validate data consistency
Recovery points	Single recovery point that likely will be compromised	Multiple recovery points
Isolation	All systems, storage and tape pools participate in the same logical system structure	Air gapped systems and storage so that logical errors and malicious intruders can not propagate
Recovery Scope	Continuous availability and disaster recovery	Forensic, surgical or catastrophic recovery capabilities



IBM Z Cyber Vault

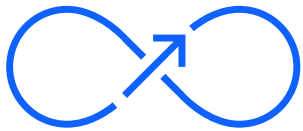


IBM Z Cyber Vault



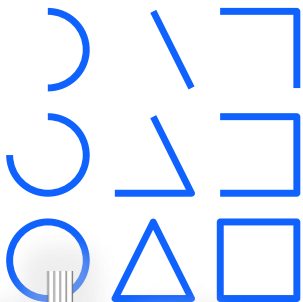
✓ IBM Z Cyber Vault Storage

- DS8000 with Safeguarded Copy
- TS7700 with LWORM Retention



✓ IBM Z Cyber Vault Automation

- GDPS Logical Corruption Protection (LCP) Manager
- IBM Technology Expert Labs Cyber Vault data validation services asset



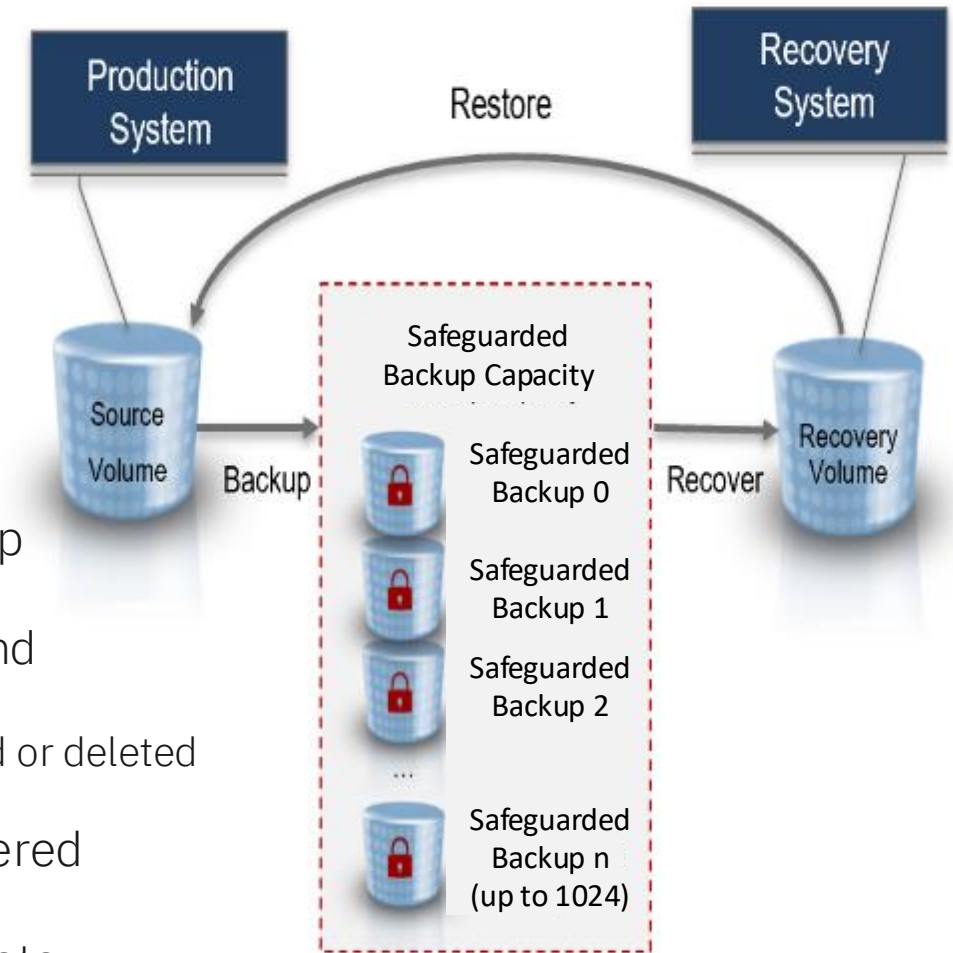
✓ IBM Z Cyber Vault Environment

- IBM Z Hardware including CPs, zIIPs, ICF, memory and required infrastructure
- IBM Z Cyber Vault Environment pid for SW licensing
- IBM Z Software Tools including, IZBR, Db2 Recovery Expert, Db2 Tools, etc.



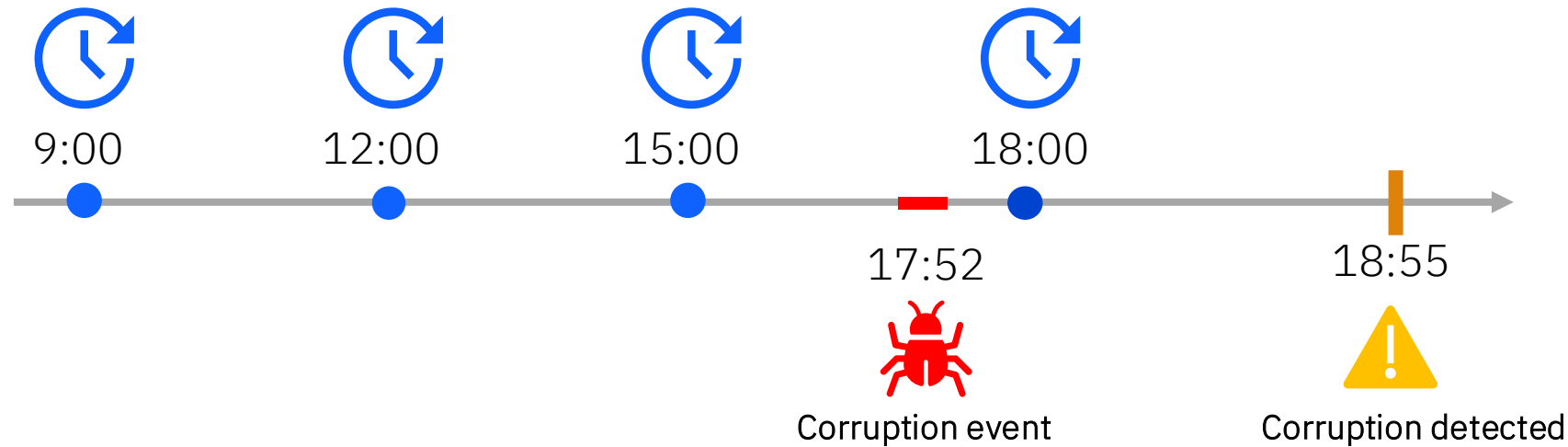
DS8K Series Safeguarded Copy

- Safeguarded Copies are secure, point-in-time copies of production data that can later be used for identification, repair, or replacement of production data that has been compromised by either Cyber or internal attack or corrupted by system failures or human error
 - Provides functionality to create up to 1024 Safeguarded Backups for a source volume
 - Source devices are where the production copies are taken from
 - Production source devices
 - HA/DR copy using data replication
- They are stored in a storage space that is called Safeguarded Backup Capacity which is hidden and not accessible by any server
 - Protection devices provide one or more logical protection copies and are not accessible by any system
 - Additional security measures aim to protect these from being modified or deleted due to user errors, malicious destruction or ransomware attacks
- The data can only be accessed after a Safeguarded Backup is recovered to a separate recovery volume
- Recovery volumes can be used with a recovery system to perform data validation, forensic analysis or to restore production data. It is also possible to restart the production systems directly on the Recovery Volume



Recovery Point Objective (RPO)

Safeguarded Copy Frequency (3 hours)



- Safeguarded Copy (SGC) is taken every 3 hours
- Corruption event initiated at 17:52
- Last good SGC prior to corruption was created at 15:00
- Corruption identified at 18:55 in the live production system
- Problem:
 - Last good SGC was taken at 15:00
 - Point-in-Time recovery would result in *2 hours and 52 minutes* of data loss

Recovery Point Objective (RPO) Roll Forward

- Additional Recovery Assets are required

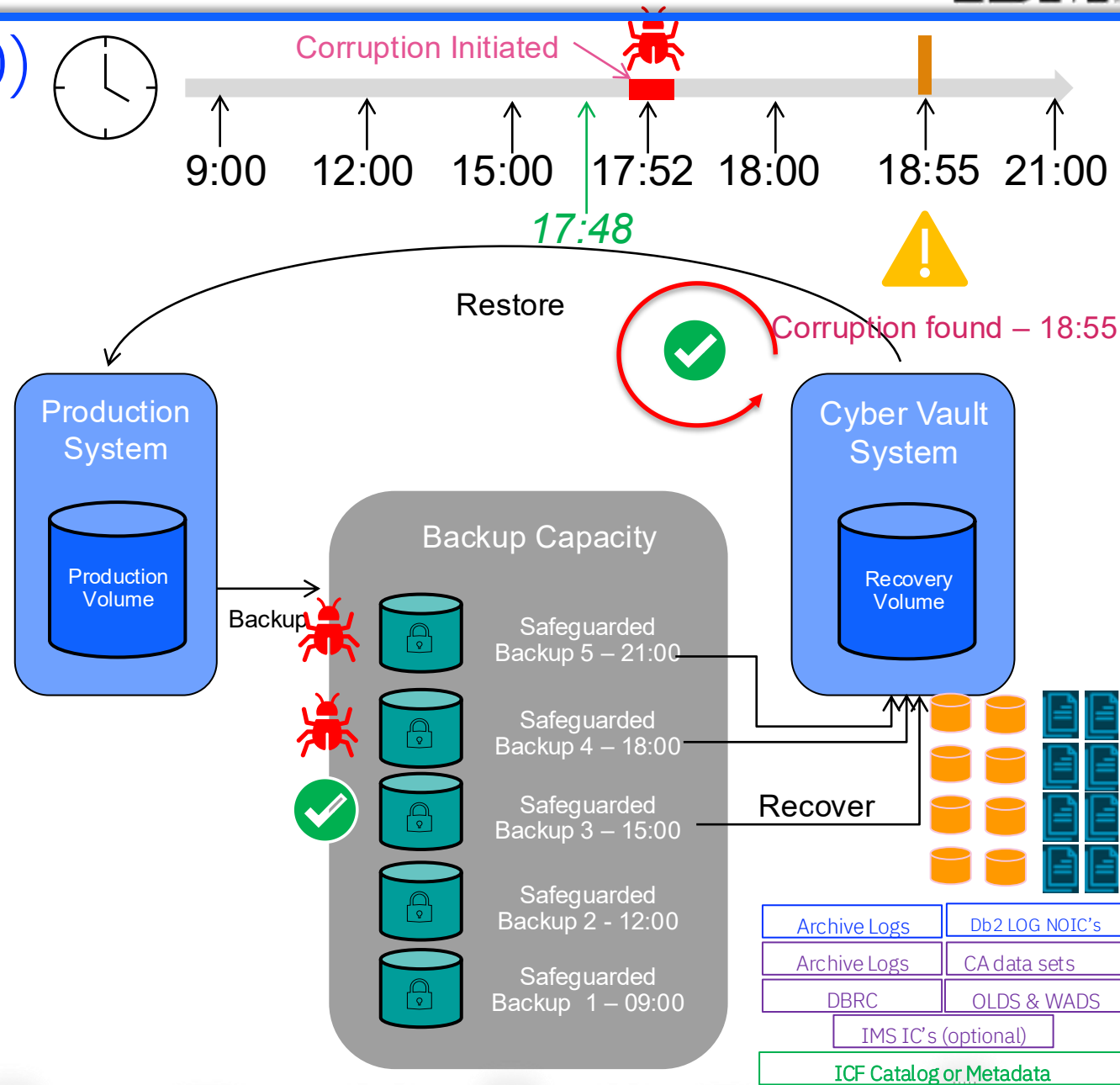
- Db2 for z/OS
 - ✓ Archive Logs
 - ✓ LOG NO Image Copies
 - ✓ ICF Catalog or Metadata
- IMS for z/OS
 - ✓ Archive logs
 - ✓ Change Accumulation data sets
 - ✓ DBRC Recon data sets
 - ✓ OLDS and WADS data sets
 - ✓ Image Copies (optionally)
 - ✓ ICF Catalog or Metadata

- Minimize IMS & Db2 data loss

- Reuse traditionally established reusable disaster recovery/group restart procedures (initial step)
- IMS – Db2 Transaction Consistent Recovery Point

- Operational steps

- The data is accessible only after a Safeguarded Backup is recovered to a separate recovery volume IPL Cyber Vault Recovery LPARs
- Db2 Group Restart
 - ✓ Roll forward
 - ✓ RESTORE LOGONLY
- IMS Recovery
- Data Validation

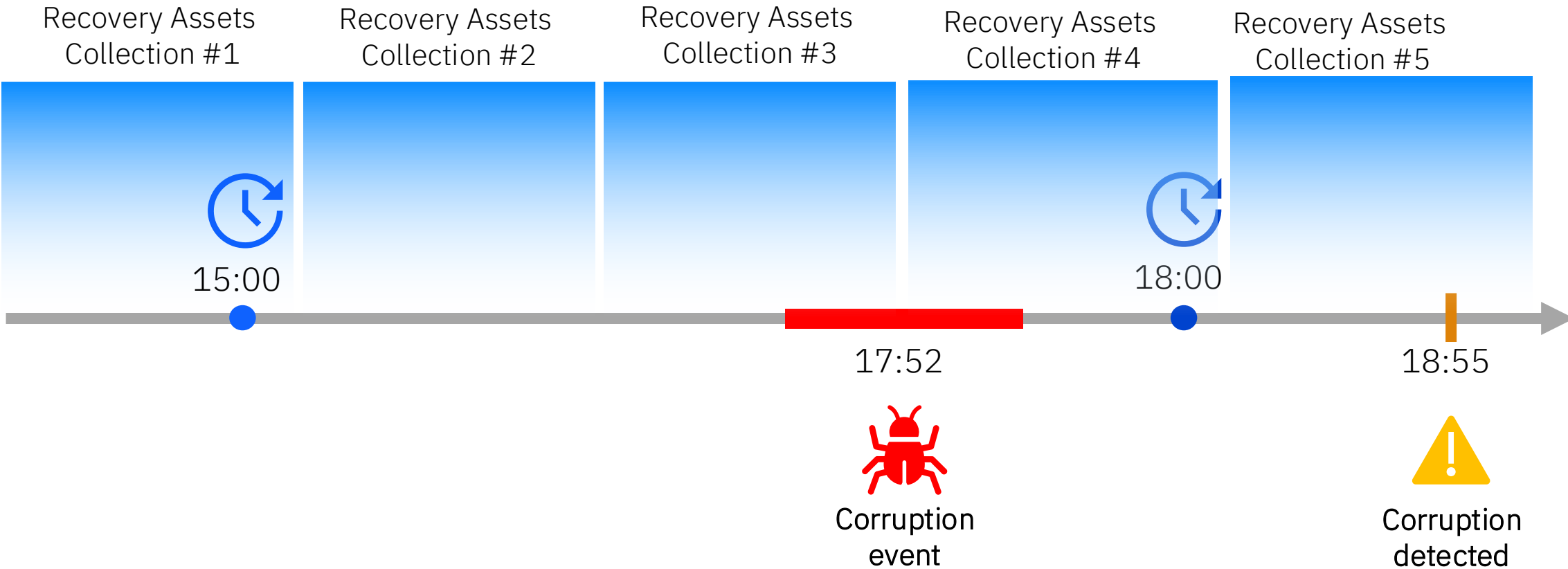


Roll Forward Recovery

- Db2 13 Recovery Roll Forward
 - Db2 13 RESTORE SYSTEM LOGONLY
 - Recovery starting point and ending point must be established
 - Recovery starting point
 - Recovery Base Log Point (RBLP) continuously being incremented
 - Recovery ending point
 - SYSPITRT
 - SYSPITR
 - Db2 13 provides additional RESTORE SYSTEM LOGONLY capabilities
 - RBLP is updated by Db2 every 5 minutes, starting point always incremented
 - RESTORE SYSTEM LOGONLY to point in time is now very appealing
 - Db2 mass application object level recovery is technically an option but for large, active systems the recovery time will likely be significantly longer than restoring an entire system with one log apply event
 - Lack of planning, intelligent design, optimization and large-scale recovery testing
 - Not optimizing job scheduling based on object size, update rate, number and size of indexes all leading to elongated recovery time
 - Not efficiently using of Db2 Fast Log Apply (FLA)
 - No prioritized list of application objects and inter-dependencies



Recovery Point Objective (RPO)



Db2 Recovery Assets: Archive Logs, Image Copies, Metadata

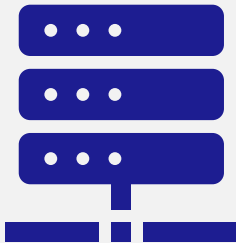
IMS Recovery Assets: Archive Logs, CA, DBRC Recons, OLDS & WADS, Image Copies, Metadata



Resource collection

Production SYSPLEX

Recover technology started task



Archive logs



Change Accum

Recons



BSDS

Image Copies



Olds & Wads

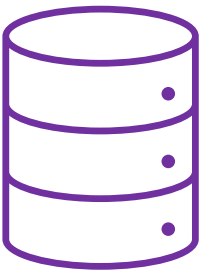


Metadata

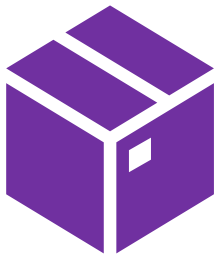
Immutable Storage



S3 Cloud Storage

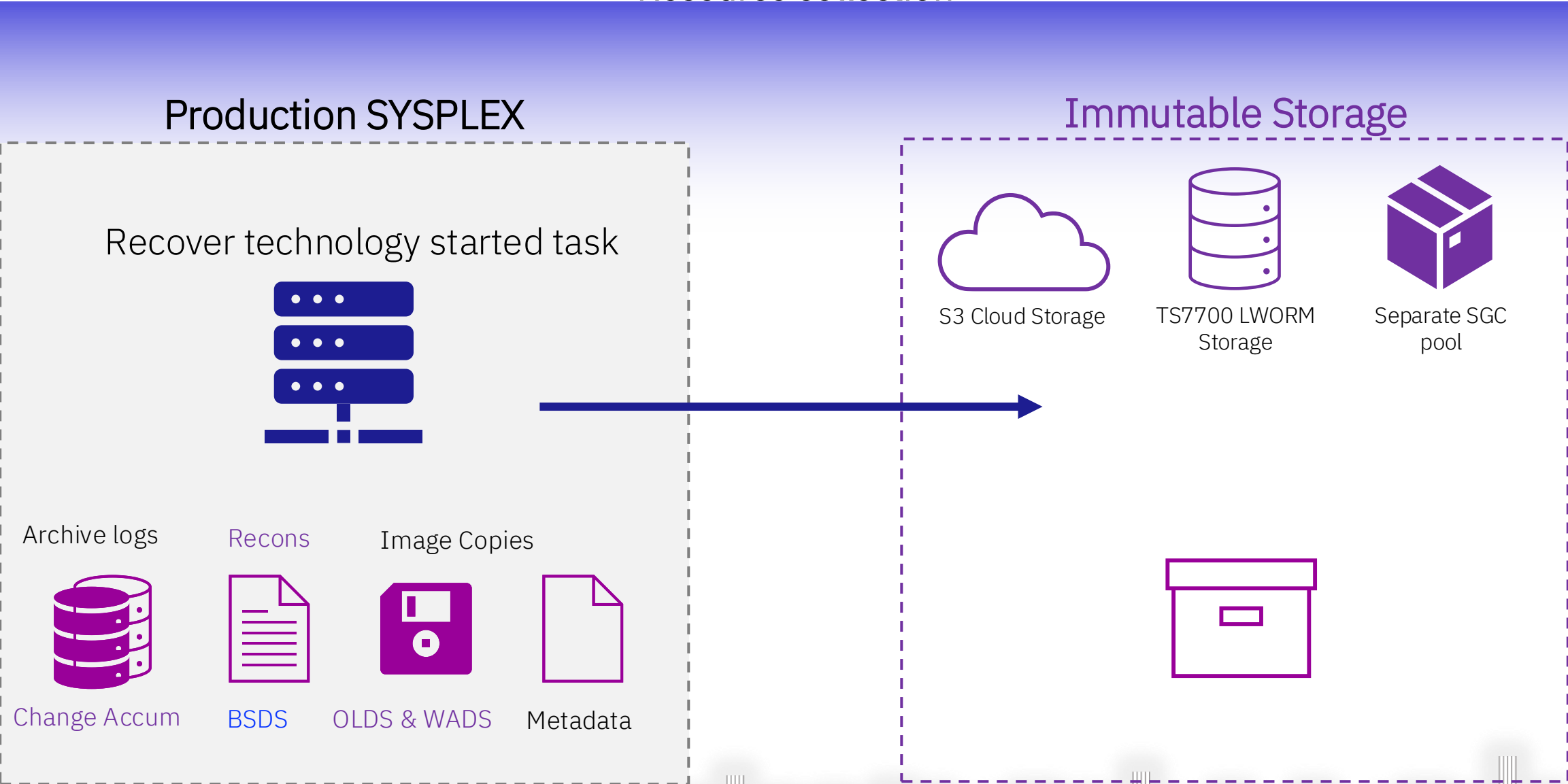


TS7700 LWORM
Storage

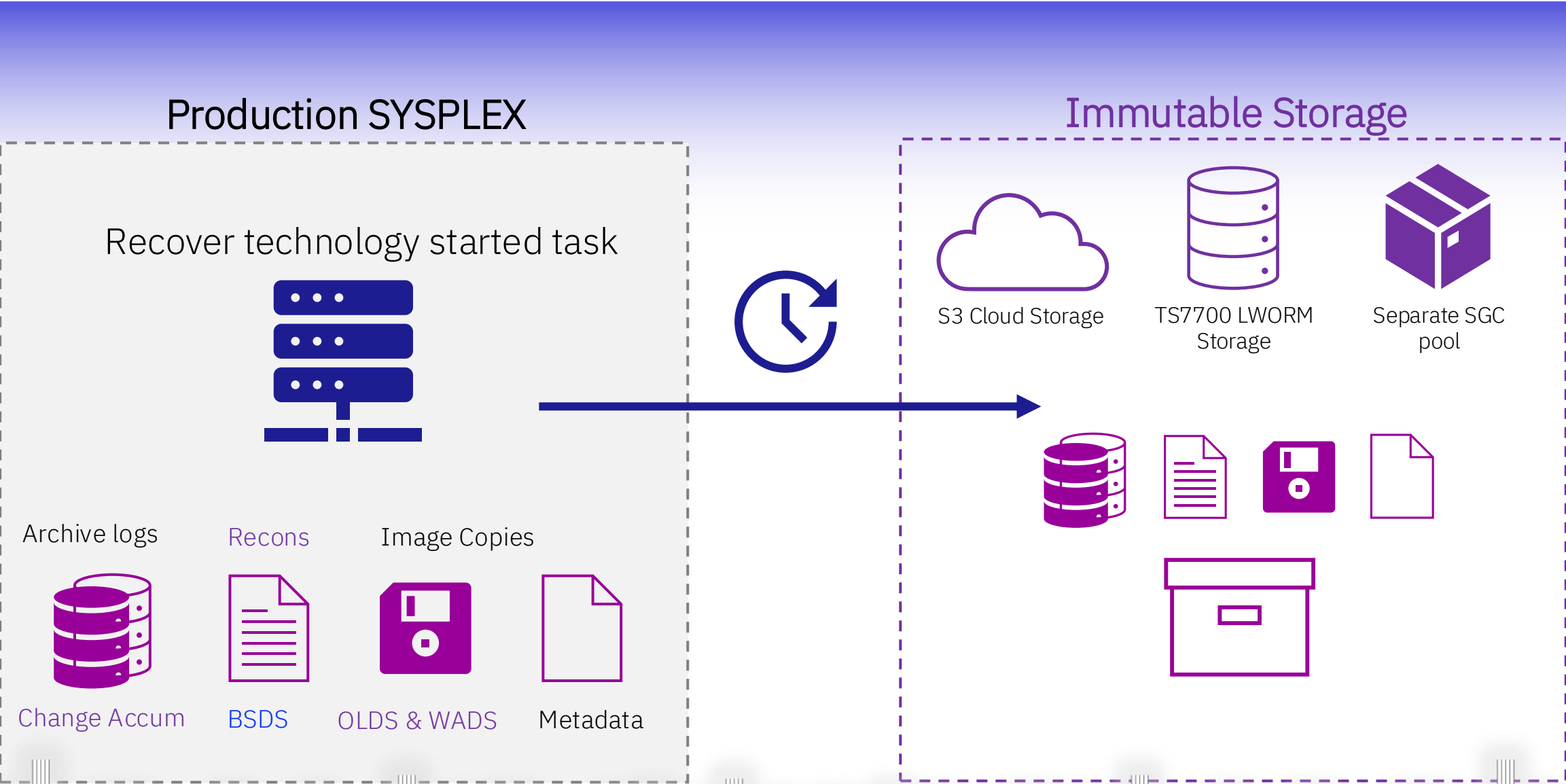


Separate SGC
pool

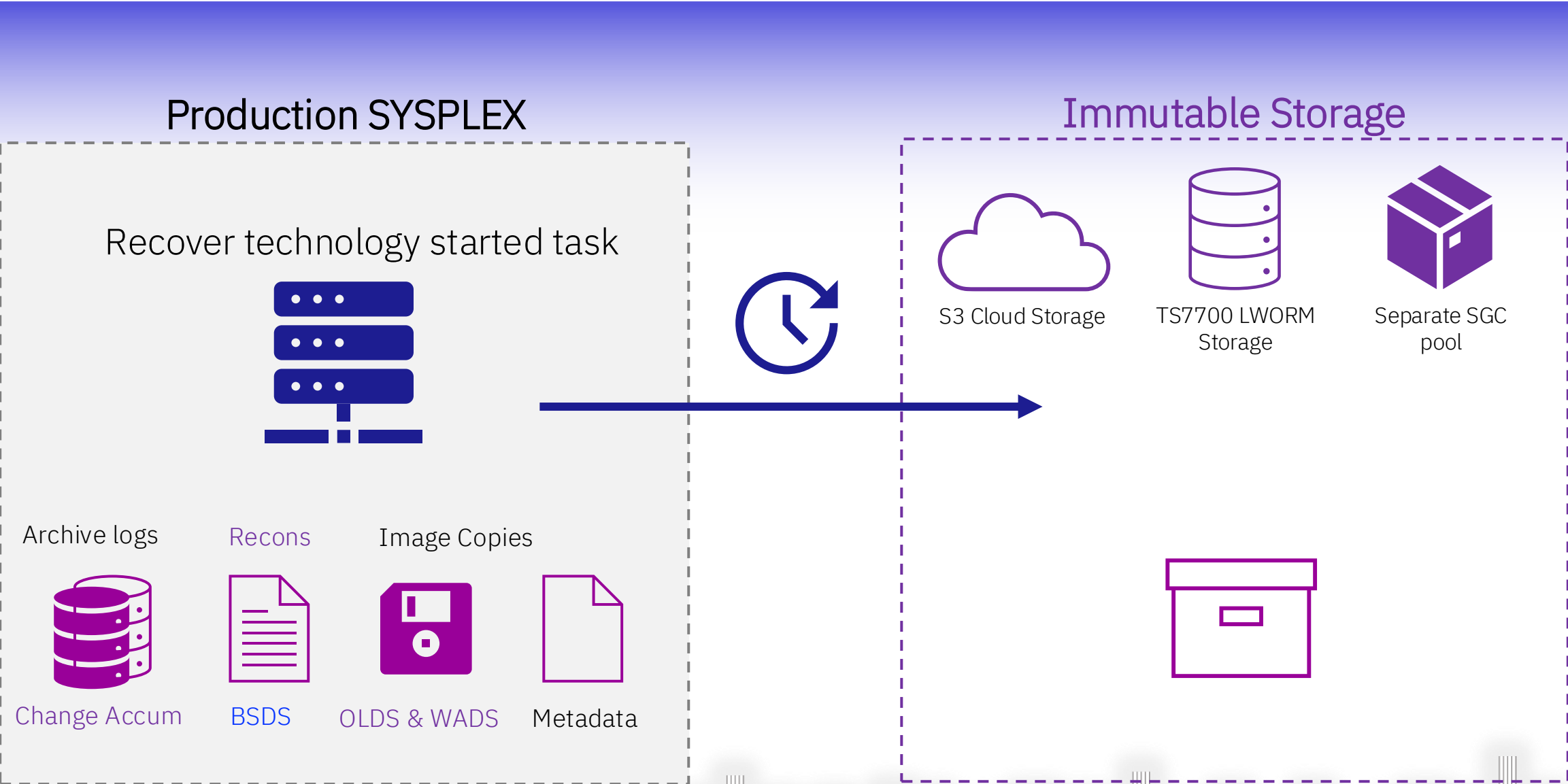
Resource collection



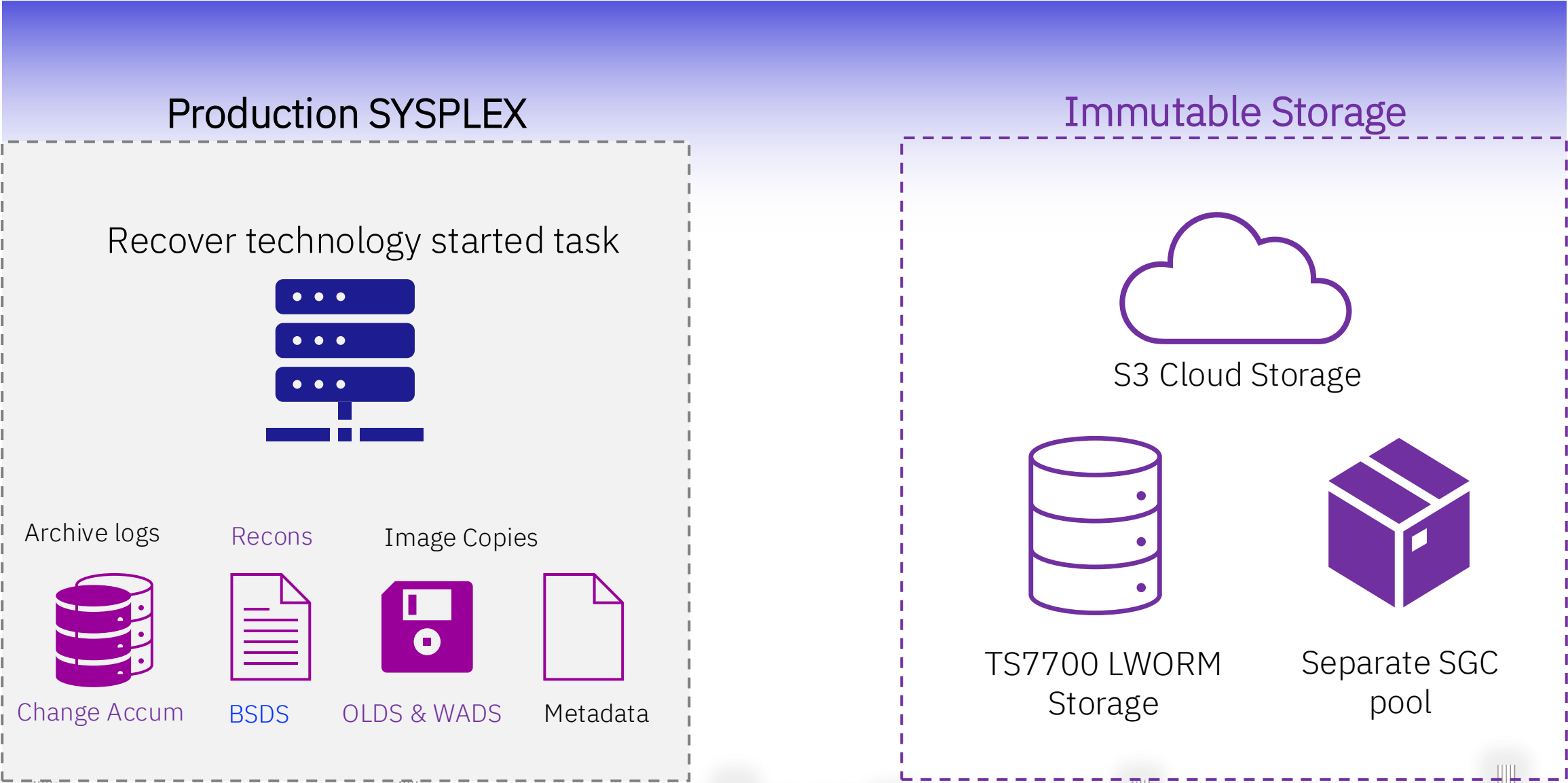
Resource collection



Resource collection



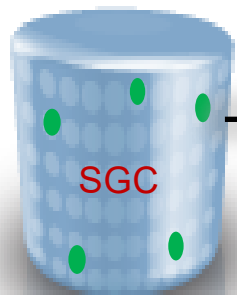
Resource collection



Improve Recovery Point Objective (RPO)

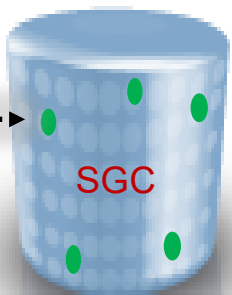
- Write additional Db2 recovery assets to immutable storage at predefined intervals (in minutes) that is accessible to Cyber Vault Recovery LPAR(s)
 - An additional 2nd Safeguarded Copy Session
 - IBM Cloud Tape Connector (S3 Cloud)
 - IBM TS7700 with LWORM Retention
- In the event of a cyber attack, user restores storage back to last good Cyber Vault copy
- Enable the access or restore the more frequently captured recovery assets in the Cyber Vault recovery LPARs
- Conditions the Db2 environment and rolls forward Db2 to just before the cyber event

Safeguarded Backup (Session 1)
every x number of *hours*



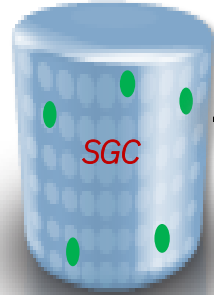
- All z/OS Storage
- Db2 for z/OS
 - ✓ Db2 Catalog & User Data
 - ✓ Db2 BSDS Copy 1
 - ✓ Db2 Active Log & Archive Logs
 - ✓ Db2 Archive Log
 - ✓ Db2 LOGNO Image Copies

Safeguarded Backup (Session 2)
every x number of *minutes*



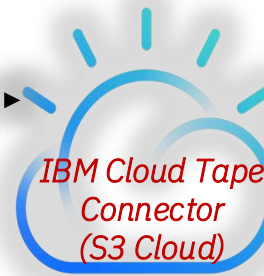
- All z/OS Storage
- Db2 for z/OS
 - ✓ Db2 Catalog & User Data
 - ✓ Db2 BSDS Copy 2
 - ✓ Db2 Active Log Copy 2
 - ✓ Db2 LOGNO Image Copies

Safeguarded Backup (Session 1)
every x number of *hours*



- All z/OS Storage
- Db2 for z/OS
 - ✓ Db2 Catalog & User Data
 - ✓ Db2 BSDS Copy 1
 - ✓ Db2 Active Log & Archive Logs
 - ✓ Db2 Archive Log
 - ✓ Db2 LOGNO Image Copies

Recovery Asset are written to IBM
Cloud Tape Connector (S3 Cloud)
every x number of *minutes*



- All z/OS Storage
- Db2 for z/OS
 - ✓ Db2 Catalog & User Data
 - ✓ Db2 BSDS Copy 1
 - ✓ Db2 Active Log & Archive Logs
 - ✓ Db2 Archive Log
 - ✓ Db2 LOGNO Image Copies

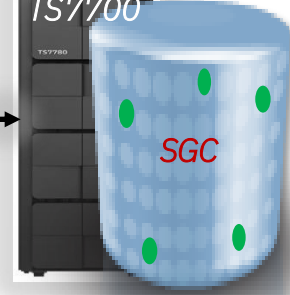
Safeguarded Backup (Session 1)
every x number of *hours*



- All z/OS Storage
- Db2 for z/OS
 - ✓ Db2 Catalog & User Data
 - ✓ Db2 BSDS Copy 1
 - ✓ Db2 Active Log & Archive Logs
 - ✓ Db2 Archive Log
 - ✓ Db2 LOGNO Image Copies
- *Tape Meta Data*

TS7700 tapes are
synchronously/asynchronously
replicated

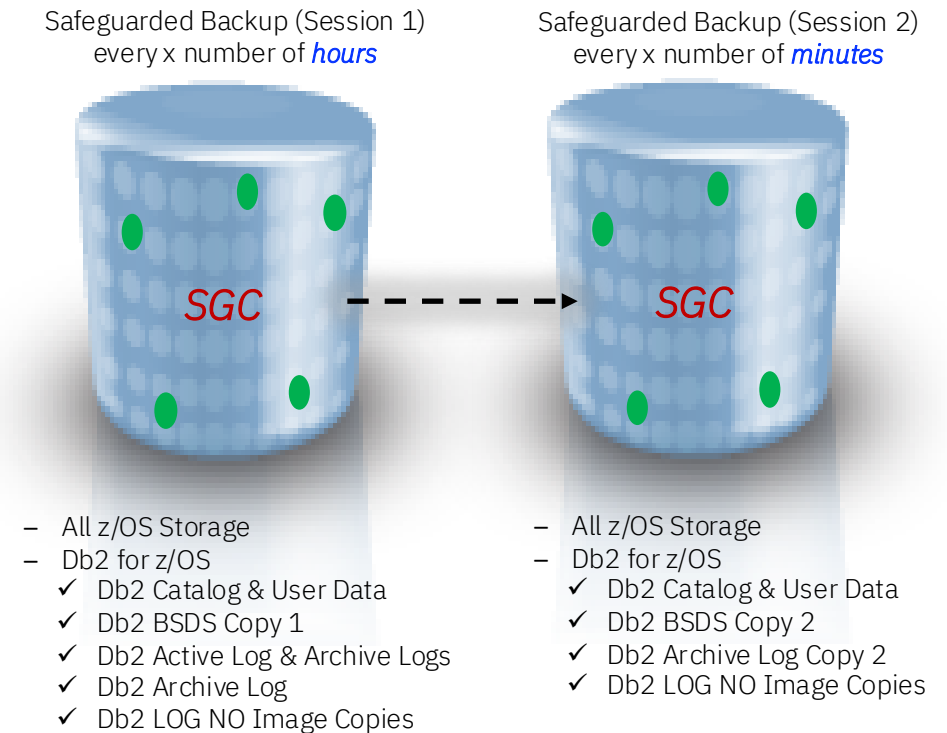
Safeguarded Backup (Session 1)
every x number of *minutes*



- All z/OS Storage
- Db2 for z/OS
 - ✓ Db2 Catalog & User Data
 - ✓ Db2 BSDS Copy 1
 - ✓ Db2 Active Log & Archive Logs
 - ✓ Db2 Archive Log
 - ✓ Db2 LOGNO Image Copies
- *Tape Meta Data*

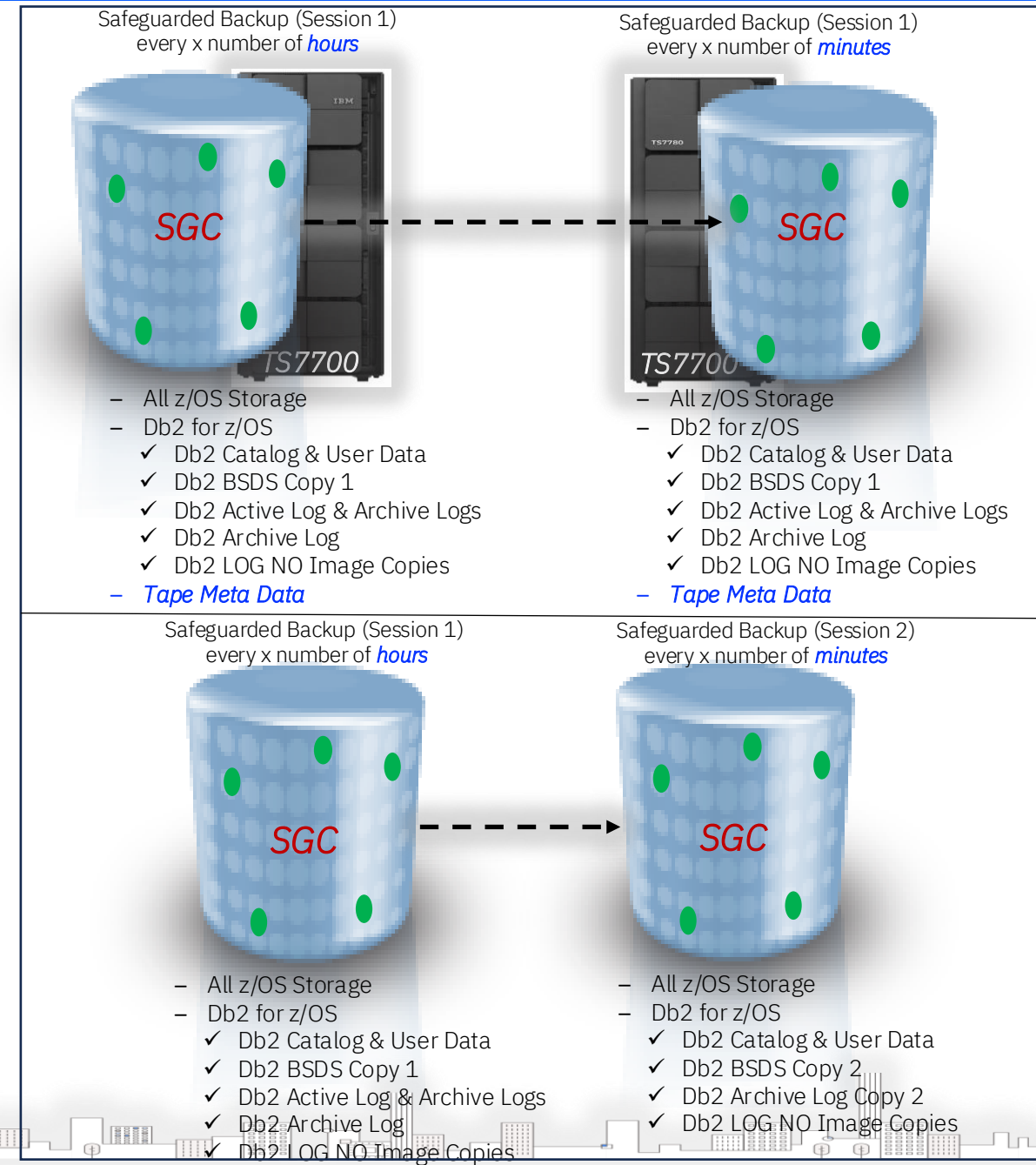
Cyber Vault Storage Options

- DS8K 2nd Safeguarded Copy Session
- Write additional Db2 recovery assets to immutable storage at predefined intervals (in minutes) that is accessible to Cyber Vault Recovery LPAR(s)
 - An additional 2nd Safeguarded Copy Session (SGC 2)
 - Db2 Recovery Assets are written to the SGC 2 volumes
 - Archive LOG Copy 2
 - Log NO Image Copy
 - ICF Catalog or metadata for additional Recovery Assets
- IBM Db2 Recovery Expert Pro v1.1
 - **APAR PH70769**
 - Validates the Archive Logs and LOG NO image copies are stored on SGC 2 volumes
 - **APAR PH71380**
 - Creates a 3rd copy of the Db2 Archive Logs and corresponding Boot Strap Data set (BSDS)
 - Creates an additional copy of any “rogue” image copy not residing on the SGC 2 volumes
 - Cyber Vault Recovery System
 - Provisions Db2 to use IBM Db2 Recovery Expert Pro 3rd copy of the archive log
 - Copy the LOG NO image copies back to original location



Cyber Vault Storage Options ...

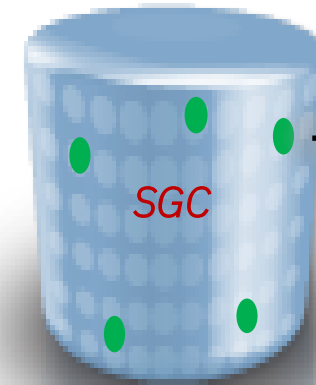
- Hybrid TS7700 with DS8K 2nd Safeguarded Copy Session
- Write additional Db2 recovery assets to immutable storage at predefined intervals (in minutes) that is accessible to Cyber Vault Recovery LPAR(s)
 - IBM TS7700 (LWORM)
 - Db2 Recovery Assets are written to TS7700 (LWORM) devices
 - Archive LOG Copy 2
 - Log NO Image Copy
 - An additional 2nd Safeguarded Copy Session (SGC 2)
 - TS7700 needed Recovery Assets
 - IBM Db2 Recovery Expert Pro metadata
 - RMM Catalog
 - TS7700 TCDB
 - ICF Catalog or metadata for additional Recovery Assets
- IBM Db2 Recovery Expert Pro v1.1
 - APAR PH70769**
 - Validates the Archive Logs and LOG NO image copies are stored on TS7700 tape devices
 - User is responsible for writing TS7700 tapes and metadata to the Cyber Vault Recovery System
 - TS7700 synchronous or asynchronous tape replication



Cyber Vault Storage Options ...

- Write additional Db2 recovery assets to immutable storage at predefined intervals (in minutes) that is accessible to Cyber Vault Recovery LPAR(s)
 - IBM Cloud Tape Connector (S3 Cloud)
- IBM Db2 Recovery Expert Pro v1.1 (at GA date)

Safeguarded Backup (Session 1)
every x number of *hours*



- All z/OS Storage
- Db2 for z/OS
 - ✓ Db2 Catalog & User Data
 - ✓ Db2 BSDS Copy 1
 - ✓ Db2 Active Log & Archive Logs
 - ✓ Db2 Archive Log
 - ✓ Db2 LOG NO Image Copies

Recovery Asset are written to IBM Cloud
Tape Connector (S3 Cloud) every x number
of *minutes*



- All z/OS Storage
- Db2 for z/OS
 - ✓ Db2 Catalog & User Data
 - ✓ Db2 BSDS Copy 1
 - ✓ Db2 Active Log & Archive Logs
 - ✓ Db2 Archive Log
 - ✓ Db2 LOG NO Image Copies

End-to-End Cyber Vault Recovery – Db2

✓ Preparation and recovery steps

Tower 1

Preparation at the production system

- ✓ Set archive log frequency based on asset collection interval
- ✓ Periodically save recovery assets to immutable storage that is accessible at the recovery system
 - ✓ Archive logs with BSDS
 - ✓ Image copies for table spaces and indexes after LOG NO events
 - ✓ ICF catalog information about the recovery assets

Tower 2

Preparation at the Cyber Vault recovery system after Safeguarded Copy is restored

- ✓ From immutable storage retrieve:
 - ✓ Archive logs, BSDS, and image copies that are more recent than the Safeguarded Copy that was restored.
- ✓ For each Db2 member:
 - ✓ Replace the BSDS with the most recent BSDS.
 - ✓ Run DSN1LOGP on most recent archive log data set.
- ✓ Identify SYSPITR system recovery point - lowest LRSN across the members from the DSN1LOGP output. This is the ending point for the roll forward (log apply).
- ✓ Allow user to override SYSPITR system recovery point

Tower 3

Recovery roll forward at the Cyber Vault recovery system

- ✓ Conditional restart process with SYSPITR
- ✓ Run RESTORE SYSTEM LOGONLY
- ✓ Stop and start Db2
- ✓ Terminate any outstanding utilities
- ✓ Run RECOVER utility on table spaces and indexes in RECOVER-pending status
- ✓ Run REBUILD INDEX on indexes in REBUILD-pending status
- ✓ Validation of data

End-to-End Cyber Vault Recovery - IMS

✓ Preparation and recovery steps

Tower 1

Preparation at the production system

- ✓ Set archive log frequency based on asset collection interval
- ✓ Periodically save recovery assets to immutable storage that is accessible at the recovery system
 - ✓ Archive logs
 - ✓ Image copies of databases, including indexes (optional)
 - ✓ Change Accumulation data sets
 - ✓ DBRC Recon data sets (IDCAMS)
 - ✓ OLDs & WADs data set images
 - ✓ ICF catalog information about the recovery assets

Tower 2

Preparation at the Cyber Vault recovery system after Safeguarded Copy is restored

- ✓ From immutable storage retrieve:
 - ✓ Archive logs SLDS
 - ✓ Image copies
 - ✓ Change accumulations that are more recent than the Safeguarded Copy that was restored
- ✓ For each IMS system :
 - ✓ Replace the Recons with the most recent Recons Images
 - ✓ Verify resource availability in DBRC vs. ICF catalog
- ✓ Identify the IMS point-in-time recovery timestamp
- ✓ Communicate to Db2 the point-in-time recovery timestamp to be converted to LRSN for Db2 SYSPITR conditional restart record

Tower 3

Recovery roll forward at the Cyber Vault recovery system (IMS)

- ✓ Run final Change Accumulation from the most current SLDS
- ✓ Recover the IMS Catalog Database
- ✓ Rebuild IMS Directory from Catalog
- ✓ Restore incremental image copies later than the last system copy (optional)
- ✓ Run Index build to "currentize" indexes
- ✓ Execute Pointer Checking to validate database
- ✓ Data validation
- ✓ Start each IMS system for access
- ✓ /ERE IMS

Recovery Point Objective Scenarios Db2 and IDAA

- IDAA Pre-Requisites for Safeguarded Copy (SGC)

- IBM DS8K and GDPS is required

- Recovery Scenario #1

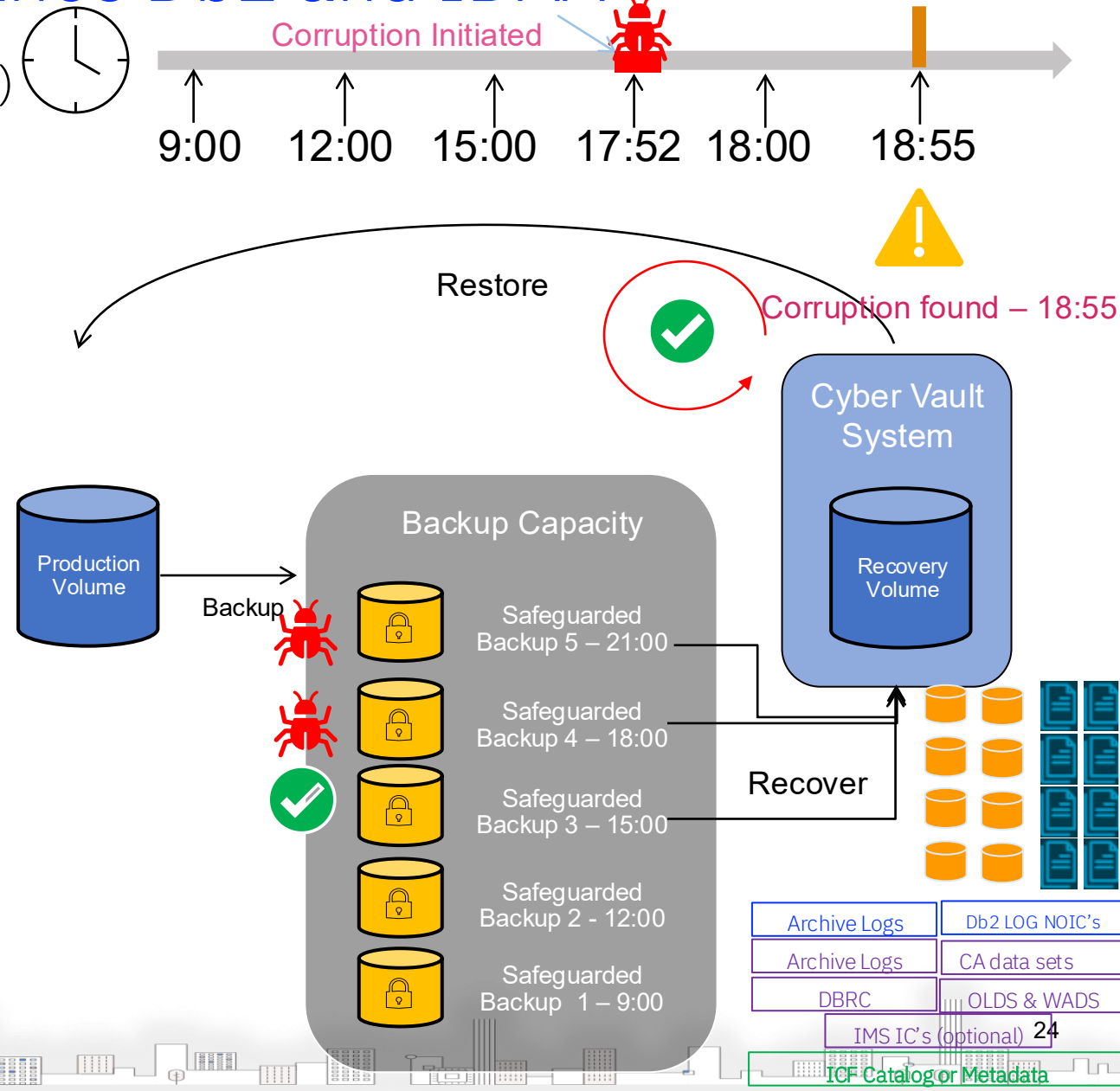
- SGC with NO Roll Forward and IDAA
 - Db2 for z/OS uses DS8K
 - IDAA uses Fibre Channel Protocol (FCP) storage

- Recovery Scenario #2

- SGC with NO Roll Forward and IDAA
 - Db2 for z/OS and IDAA uses DS8K

- Recovery Scenario #3

- SGC with Roll Forward and IDAA
 - 3a - Db2 for z/OS and IDAA uses DS8K
 - 3b - Db2 for z/OS uses DS8K, IDAA uses FCP



IDAA Cyber Vault Recovery Scenarios

Recovery Scenario #1

Storage Configuration

- ☐ Db2 for z/OS
 - IBM DS8K
- ☐ IDAA
 - Fibre Channel Protocol (FCP) storage

RPO - Data Currency Last SGC

- ☐ Db2 for z/OS
 - Point-in-Time data last SGC
- ☐ IDAA
 - Data is current as of time of corruption

RPO - Data Currency Last SGC

- ☐ Db2 and IDAA data is *not* consistent
 - IDAA is more current than Db2

Recovery Action

- ☐ All IDAA tables must be reloaded to achieve Db2 and IDAA data consistency

Recovery Scenario #2

Storage Configuration

- ☐ Db2 for z/OS & IDAA
 - IBM DS8K

RPO - Data Currency Last SGC

- ☐ Db2 for z/OS & IDAA
 - Point-in-Time data last SGC
- ☐ IDAA - Integrated Synchronization
 - Reload IDAA Tables that are enabled for replication
- ☐ Db2 & IDAA data is consistent

Recovery Action

- ☐ No action is required for tables not enabled for replication
- ☐ Reload IDAA Tables that are enabled for replication

Recovery Scenario #3

Storage Configuration

- ☐ Db2 for z/OS & IDAA
 - IBM DS8K

RPO - Data Currency Time of Corruption

- ☐ Db2 for z/OS
 - Db2 is rolled forward to the time of the corruption
- ☐ IDAA - Integrated Synchronization
 - Log Reader Task will catch up for replication enabled tables

Recovery Action

- ☐ No action is required for replication enabled tables
- ☐ Reload IDAA Tables that are not enabled for replication

Continuous Data Validation

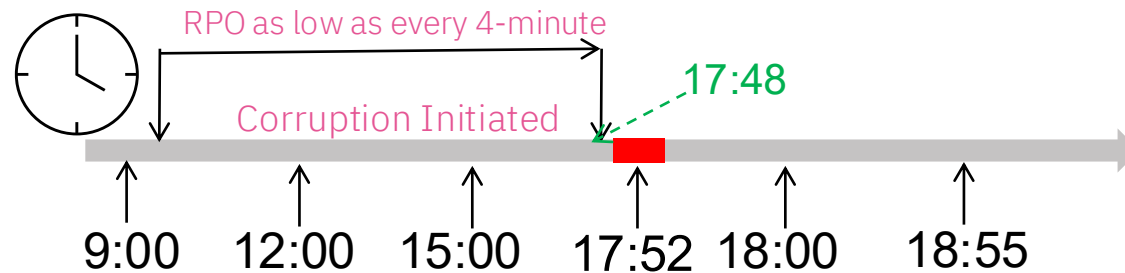
- Data validation is the process of verifying that data is clean, **accurate** and ready for use
- Frequent **continuous** automated data validation is Cyber Vault a essential activity
 - ✓ Proactive, quick identification of data corruption
 - ✓ Determine the scope of the corruption
 - ✓ Identify the date and time of the event
- **Analysis** will drive appropriate next step actions
 - ✓ Take steps to “safeguard” the production system and prevent additional and future corruption
 - ✓ Drive the data recovery plan
- Performing data validation and corruption detection against a production data clone in a recovery LPAR vs. a live production has many advantages
 - ✓ Does not impact the production system
 - ✓ Data validation procedures can be more extensive and thorough
- Valid data can be sent to offline media to have a reliable and isolated point-in-time copy

Continuous data validation allows the early detection of problems or reassurance that a given protection copy is uncorrupted.



Forensic Analysis

- Forensic analysis is the careful scientific study of the evidence to find technical facts, figure out what happened, *when* it happened and *how* the data was corrupted and/or encrypted
 - Traditionally organizations create a tertiary copy of the production environment often times well after the data has been corrupted/encrypted
 - Limits the forensic ability
- Safeguarded Copies with more granular collection of recovery assets enhances forensic analysis abilities
 - Recovery points can be as low as every 4 minutes



- Results of forensic analysis and scope determination will drive recovery actions
 - Fix the data in production
 - Extract and recover certain parts of the data from a valid backup copy (surgical recovery)
 - Point-in-time restore of the entire system on the recovery LPARs

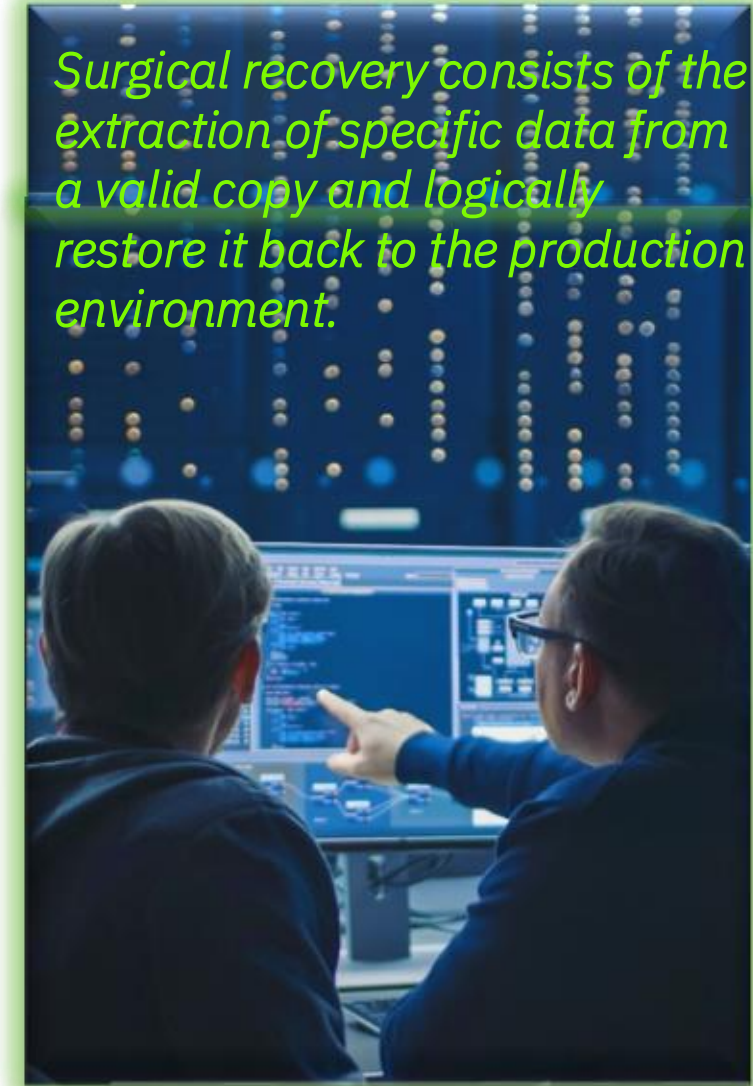
A forensic analysis identifies the cause and scope of a problem before deciding on a recovery action.



Surgical Data Recovery

- Surgical Recovery may be a faster method if only a small portion of the production data is corrupted and if consistency between current production data and the restored parts can be re-established
- Another case may occur if the last known good backup copy is too old to restore the complete environment. It may then be desirable to leave most of the production volumes in its present state, and just copy replacement data to correct actually corrupted data
- Must be super confident about understanding application and data dependencies when recovering individual datasets and subset of datasets
- Solutions designed for recovering individual dataset, subset of datasets, whole system and building a “forensic” environment need to be tested, validated and regularly practiced to make sure they are in correct working order

Surgical recovery consists of the extraction of specific data from a valid copy and logically restore it back to the production environment.



Future Db2 utilities and tools integration planned with Cyber Vault SafeGuard Copies

Questions ...



Thank You